Machine learning modeling using XGBoost and LightGBM for predicting the minimum ignition temperature of rice husk dust based on the synergistic effect of dispersion pressure and crushed brown rice

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Highlights

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Having the lowest ignition temperature at the optimal dispersion pressure.

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Crushed brown rice can improve dispersibility, leading to a decrease in MIT.

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The XGBoost model is more suitable for predicting MIT.

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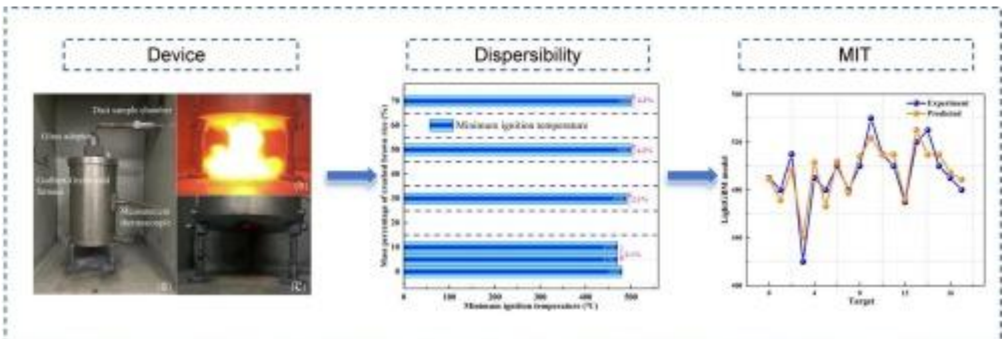
Ranking the importance of MIT using a random forest model.

Abstract

Dust explosions can occur during the production and transportation of rice husk powder. Determining the parameter of the Minimum Ignition Temperature (MIT) of dust clouds is of significant importance for mitigating dust explosions. This study investigated the influence of the addition ratio of crushed brown rice and dispersion pressure on the MIT of rice husk dust clouds. It was found that MIT exhibits a trend of initially decreasing and then increasing as dispersion pressure increases, with the lowest ignition temperature occurring at the optimal dispersion pressure. Additionally, the incorporation of large particles improves the dispersion of dust clouds, leading to a decrease in MIT and an increased risk of dust explosions. Furthermore, two machine learning modeling approaches, namely

Extreme Gradient Boosting (XGBoost) and Light Gradient Boosting Machine (LightGBM), were utilized to model the MIT of rice husk dust clouds. Hyperparameters were automatically tuned using grid search, and under optimal model conditions, the XGBoost model achieved an R^2 value of 0.911, which was significantly higher than the LightGBM model's R^2 value of 0.81, while also exhibiting lower prediction errors. This demonstrates that the XGBoost model is more suitable for studying the MIT of rice husk dust clouds. Through Random Forest importance analysis, dispersion pressure has a more pronounced impact on the MIT of rice husk dust. Therefore, when processing and transporting grain dust, these findings should be taken into account.

Graphical abstract



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Introduction

Rice husk powder, as the main by-product of the grain dust processing and transportation industry, poses a risk of dust explosion due to its manufacturing and transportation process causing suspended atmosphere [[1], [2], [3], [4]]. For example, in 2018, a grain dust explosion at the Aqaba port in Jordan tragically claimed the lives of 14 individuals [5,6]. The MIT of a dust cloud serves as a fundamental parameter in the context of dust explosions [[7], [8], [9], [10]], clarifying this parameter through experimental testing techniques holds significant importance for preventing and mitigating dust explosions.

The MIT represents the sensitivity of dust ignition and plays a pivotal role in powder transportation and processing systems [[11], [12], [13]]. The MIT is largely dependent on the size of the dust particles. Existing research has revealed that as the particle size of coal dust increases, the particles require a higher ignition energy, resulting in an exponential increase in the MIT of coal dust [14,15]. Furthermore, as the particle size of dust varies, similar phenomena in MIT are observed in aluminum dust, magnesium dust, and titanium alloy dust [[16], [17], [18]]. However, in the actual production and transportation of grain dust, fine dust particles often do not exist in isolation but coexist with larger particles, forming a mixture of coarse and fine dust. For instance, crushed brown rice (abbreviated as debris) particles with a size of approximately 450 μm, accompanied by fine rice husk dust particles with a size of only around 20 μm [5]. In our previous work, We looked into how adding varying amounts of large debris particles affected the mixed dust clouds' maximum rate of pressure rise (dP/dt)max and minimum ignition energy (MIE). We discovered that the debris particles greatly enhanced the mixed dust clouds' dispersion, raising the risk of dust ignition and the intensity of the explosion's aftereffects [5]. However, the focus of this study is to investigate the MIT of rice husk powder dust clouds under the common conditions of crushed brown rice particles mixing in the actual process of grain production. Based on the existing research is based on the fine dust, it remains unknown whether the inclusion of large debris particles will alter the MIT of rice husk dust clouds. Certainly, this phenomenon, which poses a potential danger to the grain dust processing industries, has not garnered sufficient attention from us and necessitates further investigation. Additionally, the dispersion pressure also has a notable impact on the MIT [15]. Existing research indicates that as the dispersion pressure increases, the MIT of oil shale exhibits a trend of initially decreasing and then increasing. Higher dispersion pressures shorten the residence time of oil shale dust in the furnace, as well as the heat exchange time between particles and the furnace environment, and among particles themselves [19]. Moreover, as the dispersion pressure fluctuates, similar phenomena in MIT are evident in both aluminum dust and iron dust [[20], [21], [22]]. However, the MIT of rice husk dust combined with large particles under different dispersion pressures remains unknown, and further research is needed.

Another issue that warrants our attention is that dust explosion experiments are statistical phenomena, and the experimental results may vary with changes in environmental factors, equipment errors, and testing procedures [[23], [24], [25]], furthermore, the experimental process is often costly and cumbersome [[26], [27], [28]]. Therefore, it becomes imperative to select an auxiliary tool that can provide guidance in the field of dust explosions. The introduction of several machine learning-based data-driven modeling approaches in recent years has expanded the possibilities for dust explosion predictive modeling [29,30]. Existing research has validated the applicability of models such as Quantitative Structure-Property Relationship (QSPR), Random Forest (RF), Artificial Neural Networks (ANN), and Back Propagation Neural Network (BPNN) for predicting the Minimum Ignition Energy (MIE) of dust clouds [[31], [32], [33]]. Furthermore, existing study has validated the feasibility of machine learning prediction models for various aspects such as the Lower Flammable Limit (LFL) [[34], [35], [36]], Upper Flammable Limit [37,38], and Autoignition Temperature [39,40]. Additionally, it has confirmed that modeling with Artificial Neural Networks (ANN) is feasible for forecasting the MIT of dust clouds [20,22]. The applicability of machine learning in predicting flame propagation speed has also been reported [41]. In summary, the aforementioned models possess considerable predictive value across various domains. However, considering the synergistic effect of dispersion pressure and the addition of debris content in dust clouds, as modeled through machine learning data-driven approaches in this paper, this represents a novel experimental methodology. Moreover, due to the small number of experimental groups in this study, whether machine learning models are suitable for predicting the MIT of dust clouds under these conditions remains uncertain. The purpose of our work is to find a suitable machine learning model to achieve the accurate prediction of rice husk dust cloud MIT. XGBoost is the open source framework of gradient promotion, has high efficiency and flexibility, robustness, can be autonomous learning [41]. LightGBM is a kind of lightweight gradient promotion algorithm, algorithm at the depth limit, the two algorithms for small data are showed good prediction performance [42]. Our research emphasizes the comparison of XGBoost and LightGBM machine learning models to accurately predict the MIT of rice husk dust clouds.

This project conducts experimental research on the MIT of rice husk dust clouds under the synergistic effects of different dispersion pressures and debris content, aiming to reveal the underlying influence patterns. Furthermore, based on the aforementioned experimental analysis, data modeling of the experimental data was conducted using XGBoost and LightGBM. The objective of this work is to identify a machine learning model that can accurately predict the MIT of rice husk dust clouds under these conditions by comparing the interpretability and predictive accuracy among the proposed models. The noteworthy significance of this research lies in supplying reference value and corresponding predictive tools for the study of rice husk dust explosions in grain transportation industrial processes, enabling individuals to discover methods to mitigate the damage caused by such accidents in process industries.

Experimental materials and device

Rice husk powder, due to its high frequency of accidents in practical production, was preferentially chosen as the experimental material [[43], [44], [45]]. The median diameter (D_{50}) of the rice husk powder samples is 17.56 μm (Fig. 1a). Randomly select 100 debris grains. The particle size statistics were performed for the samples and find that the particle size distribution of large grains is roughly uniform (Fig. 1b). More descriptions of the samples have been reported in previous articles [5

Effect of dispersion pressure on rice husk dust cloud MIT

Fig. 7 illustrates the MIT variations in selected rice husk powders of differing qualities, as well as those mixed with different ratios of debris. From the perspective of dispersion pressure, the MIT exhibits a similar pattern of variation. It demonstrates a trend of falling at first, then rising as the dispersion pressure increases. This aligns with the findings of existing research [[53], [54], [55]]. Furthermore, the MIT attains its minimum value under optimal dispersion pressure. For

Conclusion

This study explored the impact of large debris particle content and dispersion pressure on the MIT by mixing varying amounts of large debris particles. It was found that MIT exhibits a trend of initially decreasing and then increasing as dispersion pressure increases, with the smallest MIT occurring at the optimal dispersion pressure.Furthermore, the addition of trace amounts of large particles results in a slight decrease in the MIT of dust clouds. According to previous research, large

CRediT authorship contribution statement

Jinglin Zhang: Formal analysis. **Gang Li:** Data curation. **Zhenguo Du:** Resources. **Shikai Bao:** Formal analysis. **Chang Li:** Validation. **Xiumei Cao:** Project administration. **Chunmiao Yuan:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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